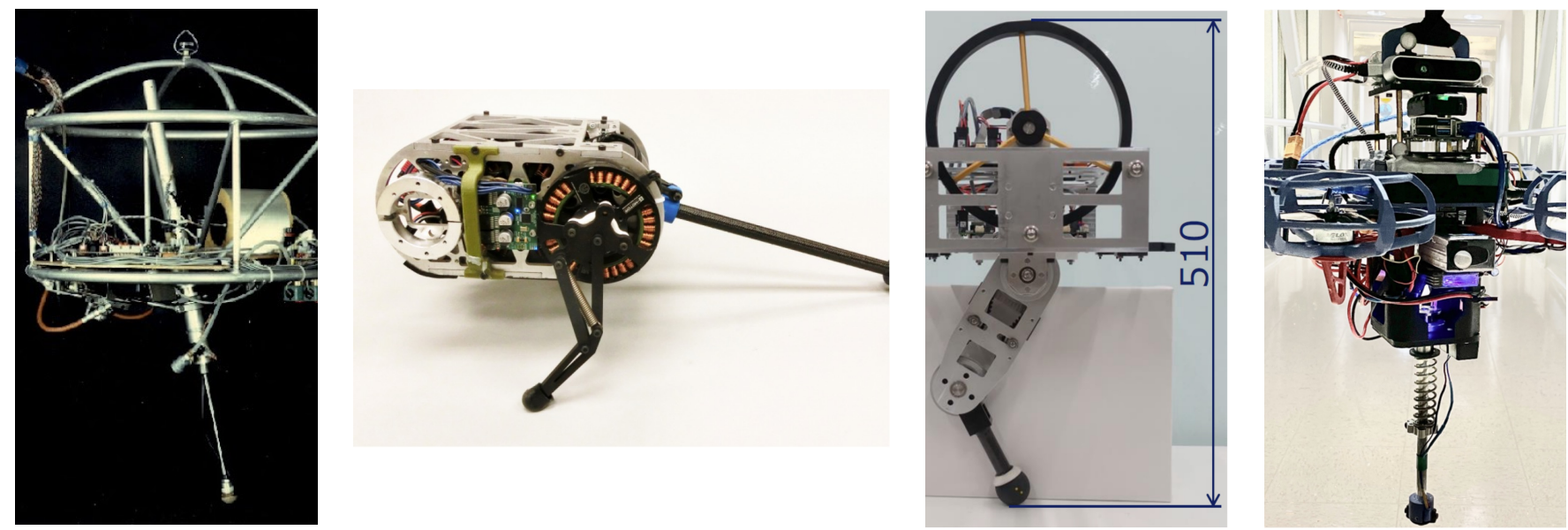


1. Problem - hybrid dynamics create an authority gap



Conventional Hopper → Inertial Appendages → Propeller Assistance

Stance

Ground reaction force exists only during a short contact interval.

Flight

No contact: attitude becomes underactuated.

Disturbance

Impacts and pushes inject attitude and horizontal momentum.

Control authority disappears exactly when attitude can keep drifting.

Design question: how should leg contact force and rotor thrust be coordinated across stance and flight?

2. First proof - stable on 30° grass



Propellers off

After asymmetric contact, attitude error grows. With no flight-phase actuator, the robot falls.

Propellers on

Differential thrust remains available through flight and restores attitude before the next landing.

What this establishes

Residual authority in stance.

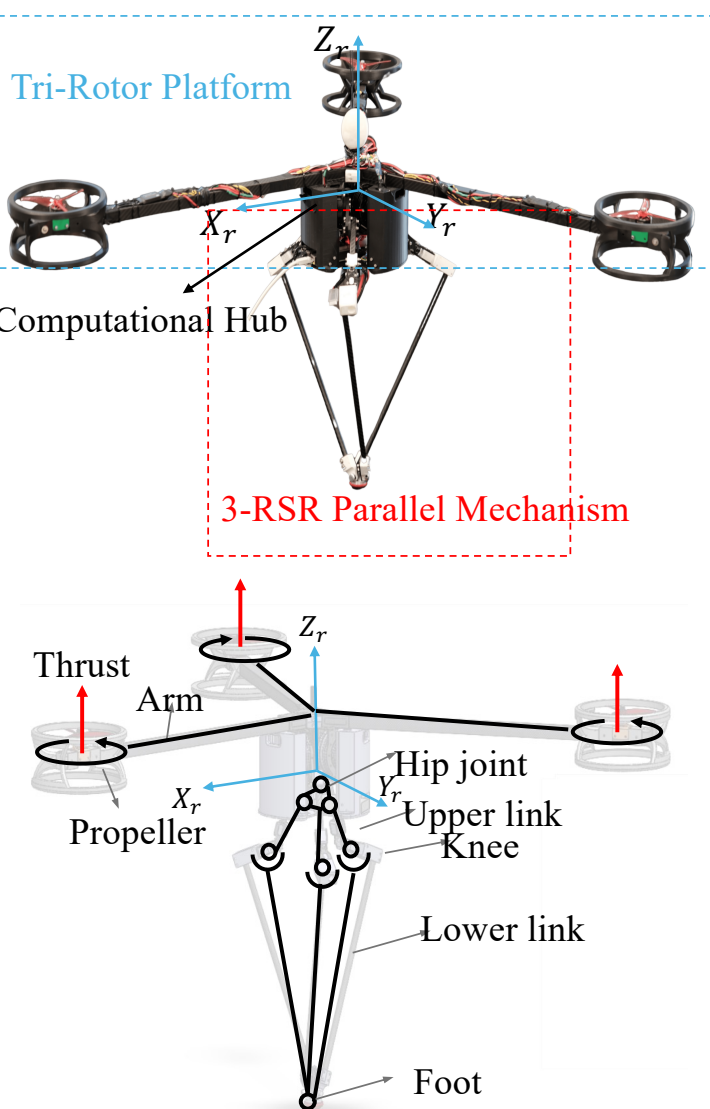
Primary attitude authority in flight.

Faster recovery from uneven impacts.

Not primary lift: the rotors close the attitude-authority gap.

Propellers off: attitude diverges and the robot falls. Propellers on: it remains upright and keeps hopping.

3. Platform - hardware built for leg-first hopping



3-RSR parallel leg

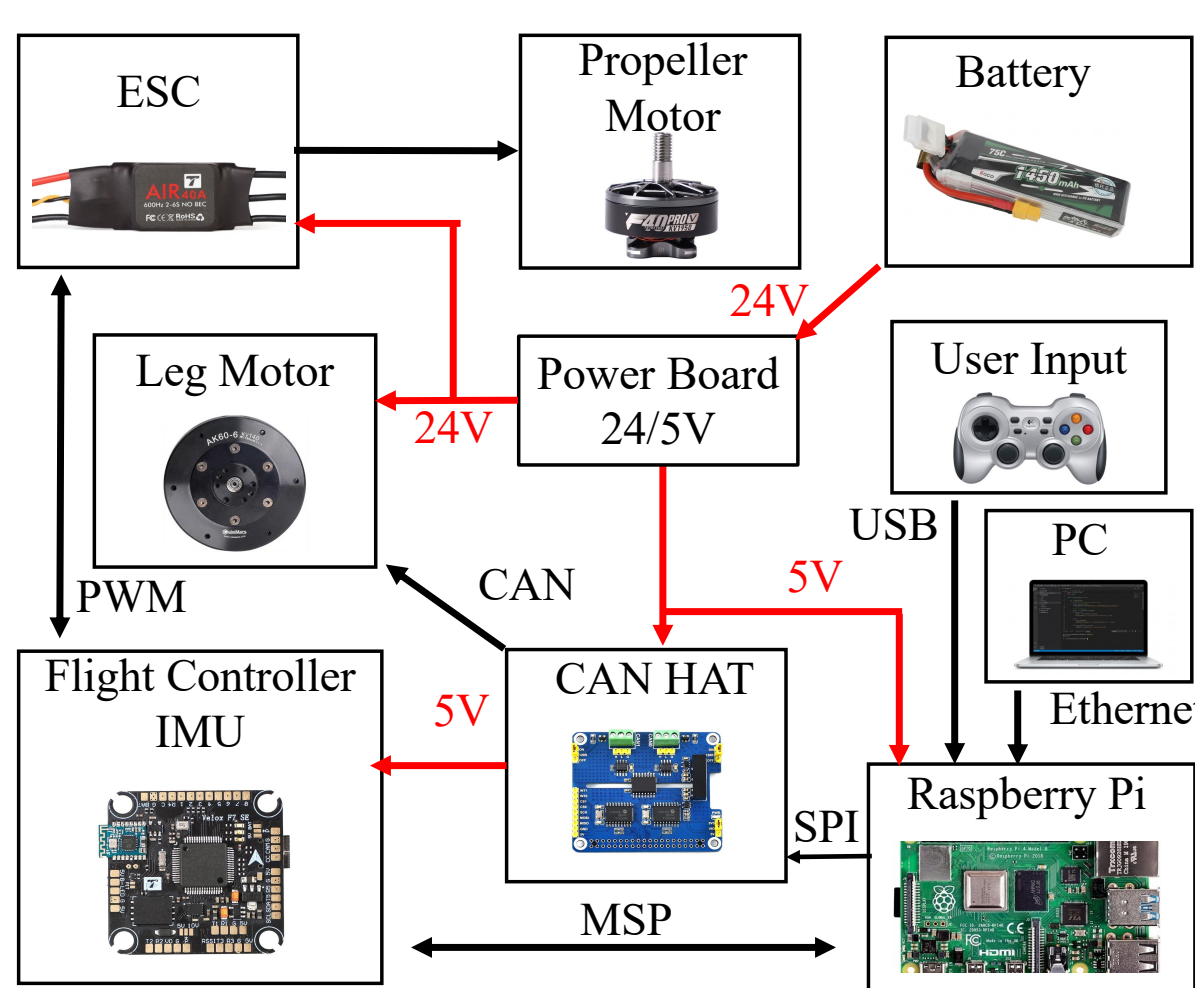
Three motor-driven RSR chains meet at one foot: fast 3-DoF omnidirectional foot placement with only **0.35 kg** moving leg mass (9.2% of the robot).

Computational hub

Batteries and electronics sit near the hip, lowering the center of mass and reducing rotational inertia.

Y-layout tri-rotor

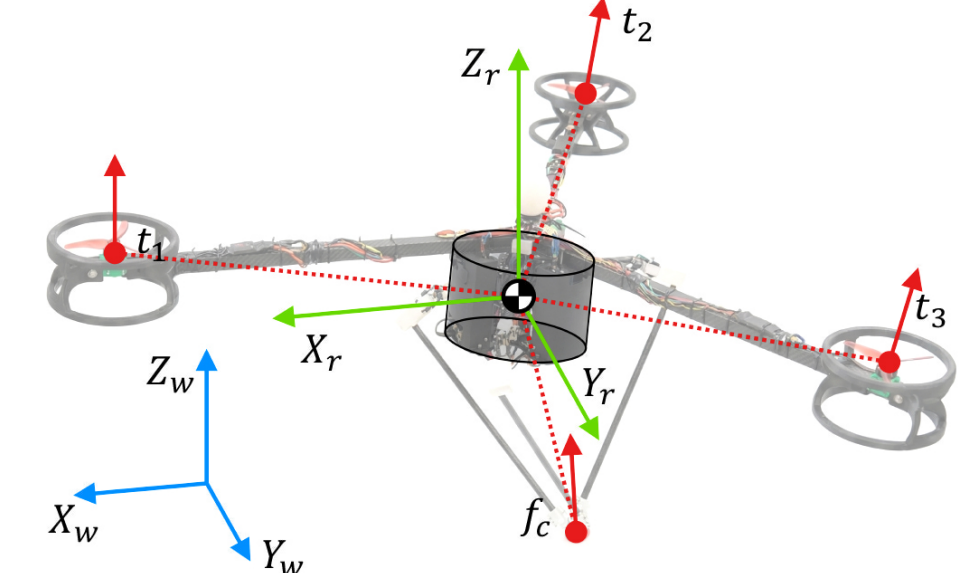
Differential thrust gives continuous attitude authority. Each rotor reaches **10 N**, but total thrust stays at 0.82 mg : a hopper, not a drone.



Paper Fig. 3 - electronic architecture.

3.75 kg total ■ 9.2% leg mass ■ 20 Nm / leg motor ■ 500 Hz control

4. Model - one rigid body, two actuator sets



Net wrench = leg contact + three rotors

$$\begin{bmatrix} \mathbf{F}_{net} \\ \boldsymbol{\tau}_{net} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_c \\ \mathbf{r}_c \times \mathbf{f}_c \end{bmatrix} + \sum_{i=1}^3 \begin{bmatrix} \mathbf{d}_i \\ \mathbf{r}_i \times \mathbf{d}_i \end{bmatrix} \mathbf{t}_i$$

Newton-Euler dynamics of the CoM

$$m\mathbf{a} = \mathbf{F}_{net} + m\mathbf{g}, \quad \dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}(\mathbf{R}^T \boldsymbol{\tau}_{net} - \boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega})$$

Paper Fig. 4 - contact force $\mathbf{f}_c$ and thrusts $\mathbf{t}_{1,2,3}$ jointly produce the body wrench.

Symbols (values used in all experiments)

$\mathbf{f}_c$ ground reaction force (world)

$\mathbf{r}_c, \mathbf{r}_i$ arms: CoM $\rightarrow$ contact, rotor i

$\mathbf{t}_i \in [0, 10]\text{ N}$ thrust along $\mathbf{d}_i = \mathbf{R}\mathbf{e}_3$

$\mathbf{F}_{net}, \boldsymbol{\tau}_{net}$ net force, moment

$m=3.75\text{ kg}$; $\mathbf{g}$ gravity; $\mathbf{I}$ inertia

$\mathbf{R} \in \text{SO}(3)$ attitude; $\boldsymbol{\omega}$ body rate

$\mathbf{J}$ leg Jacobian; $\tau_{max}=20\text{ Nm}$ limit

$\mu=0.4$ friction coefficient

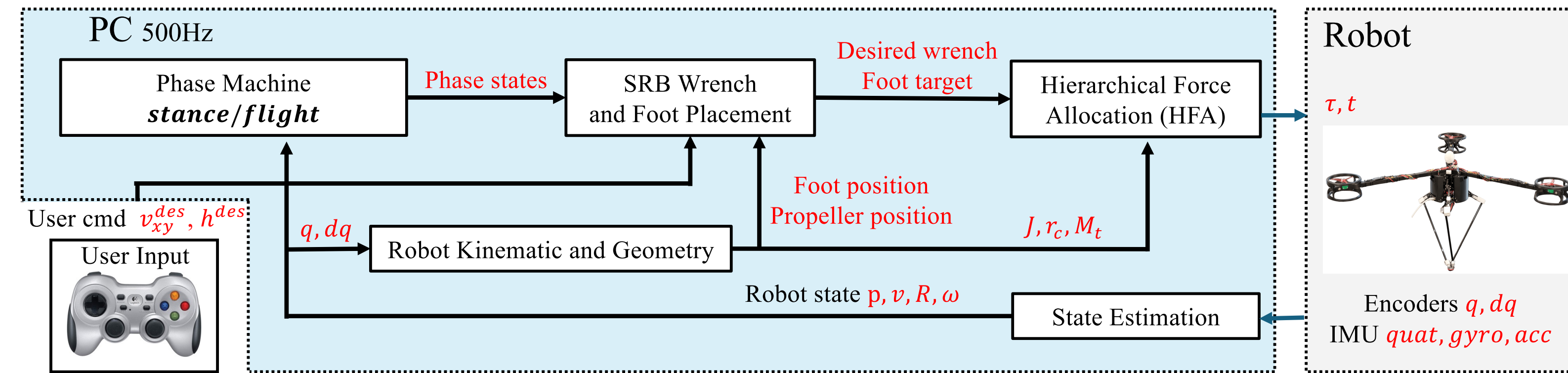
$p_z, \dot{p}_z$ CoM height, velocity; $h_{des}=0.664\text{ m}$

l leg length; $l_0=0.464\text{ m}$; Δ_{td} threshold

k_z, b_z, k_E vertical, k_v, k_r Raibert gains

$\mathbf{K}_R, \mathbf{K}_\omega$ attitude PD gains

5. Control - leg-priority hierarchical force allocation

Paper Fig. 5 - implemented 500 Hz architecture: user commands and feedback $\rightarrow$ planner $\rightarrow$ two-stage allocator $\rightarrow$ leg torques + rotor thrusts.

A. Vertical energy hopping (stance)

$$E_{sys} = \frac{1}{2}m\dot{p}_z^2 + mgp_z, \quad E_{des} = mg h_{des}$$

$$\mathbf{f}_{c,z}^{des} = \max(0, k_z(h_{des} - p_z) - b_z\dot{p}_z + k_E(E_{des} - E_{sys}))$$

$$\text{Injects push-off energy, absorbs impact.}$$

B. SO(3) attitude PD (both phases)

$$\mathbf{e}_R = \frac{1}{2}(\mathbf{R}_{des}^T \mathbf{R} - \mathbf{R}^T \mathbf{R}_{des})^\vee$$

$$\boldsymbol{\tau}_b^{des} = -\mathbf{K}_R \mathbf{e}_R + \mathbf{K}_\omega(\boldsymbol{\omega}^{des} - \boldsymbol{\omega})$$

$$\text{Roll/pitch only; yaw left free. } \boldsymbol{\tau}^{des} = \mathbf{R}\boldsymbol{\tau}_b^{des}.$$

C. Leg first - constrained least squares

$$\mathbf{f}_{c,xy}^{cmd} = \arg \min_{\mathbf{f}_{xy}} \|(\mathbf{r}_c \times \mathbf{f}_c)_{rp} - (\boldsymbol{\tau}^{des})_{rp}\|_2^2$$

$$\text{s.t. } \|\mathbf{f}_{xy}\|_2 \leq \mu \mathbf{f}_{c,z}^{des}, \quad \|\mathbf{J}^T(-\mathbf{R}^T \mathbf{f}_c)\|_\infty \leq \tau_{max}$$

The leg realizes all it can under friction and torque limits.

D. Rotors take only the residual

$$\boldsymbol{\tau}_{res}^{cmd} = \boldsymbol{\tau}^{des} - \mathbf{r}_c \times \mathbf{f}_c^{cmd}$$

$$\delta \mathbf{t}^{cmd} = \arg \min_{\delta \mathbf{t}} \|(\mathbf{M}_t \delta \mathbf{t})_{rp} - (\boldsymbol{\tau}_{res}^{cmd})_{rp}\|_2^2$$

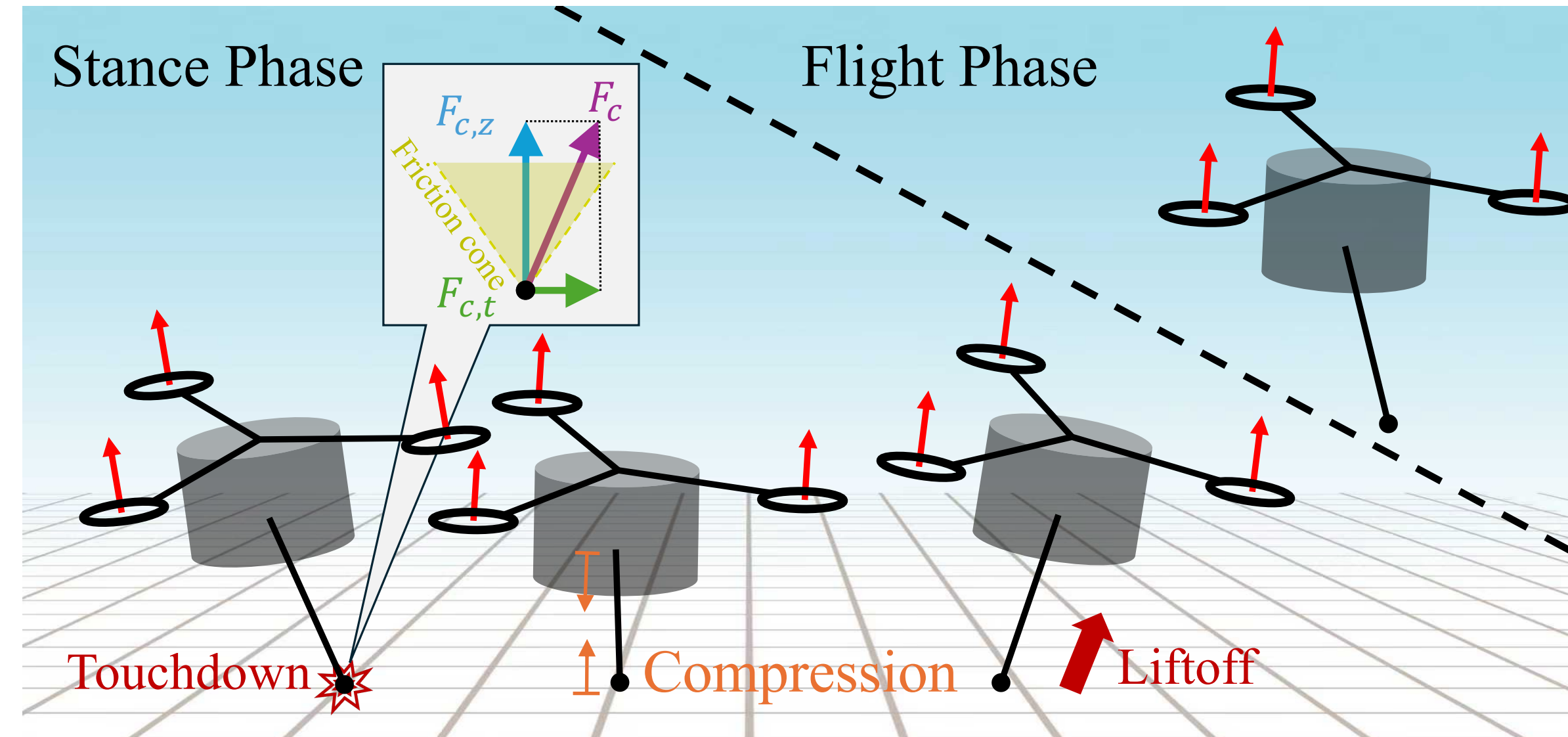
s.t. thrust bounds; $\mathbf{t}^{cmd} = \frac{T_{base}}{3}\mathbf{1}_3 + \delta \mathbf{t}^{cmd}$, $T_{base} = 0.02\text{ mg}$.

E. Flight - rotors primary for attitude; Raibert foot placement

$$\mathbf{f}_c = 0; \quad \text{rotors track } (\boldsymbol{\tau}^{des})_{rp}; \quad \text{leg tracks } \mathbf{r}_{td,xy}^{des} = k_v \mathbf{v}_{xy} - k_r \mathbf{v}_{xy}^{des}$$

By construction the rotors can only add to - never replace - the leg's contact action.

6. Onehop - event-driven authority handoff



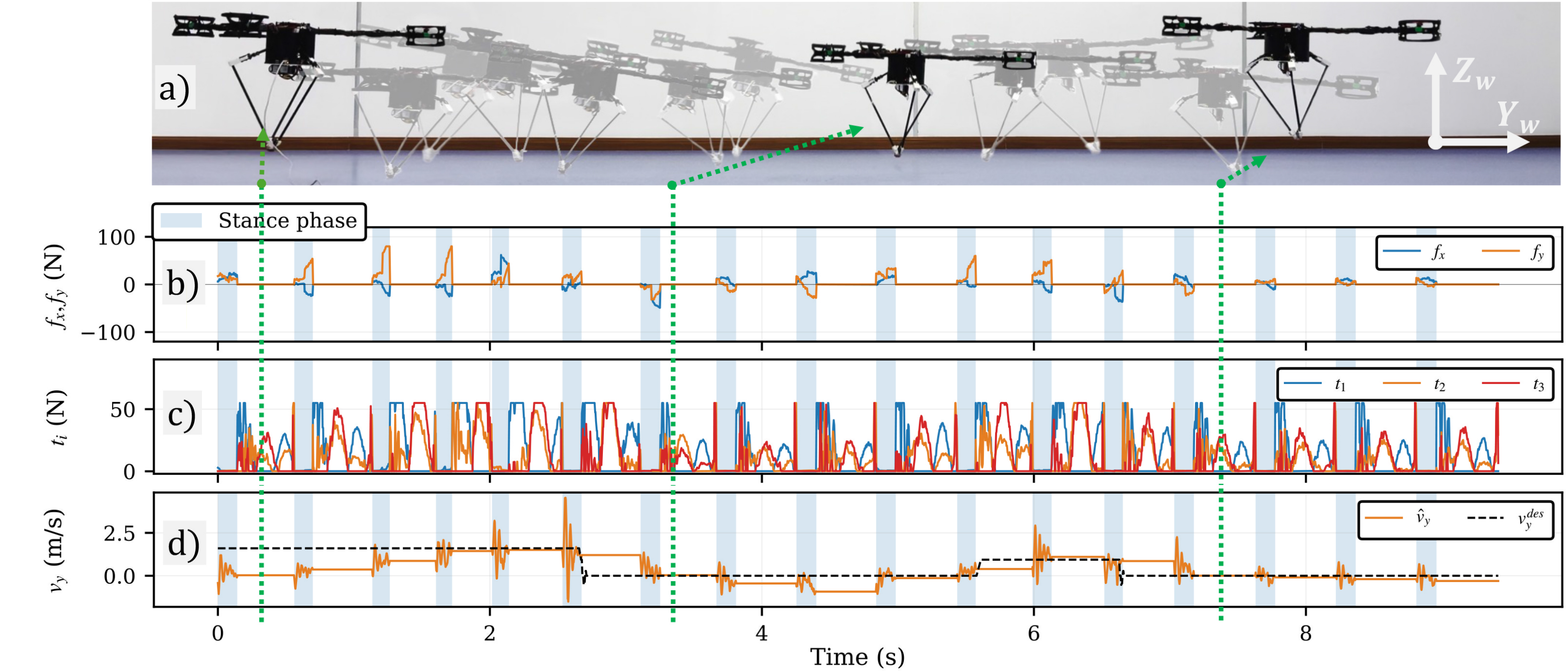
Leg authority exists only in stance; rotor attitude authority is continuous through the full hop.

Touchdown: $l \leq l_0 - \Delta_{td}$
Leg absorbs impact; energy law drives the next apex.

Liftoff: $l \geq l_0$
Contact wrench vanishes; rotors primary for attitude.

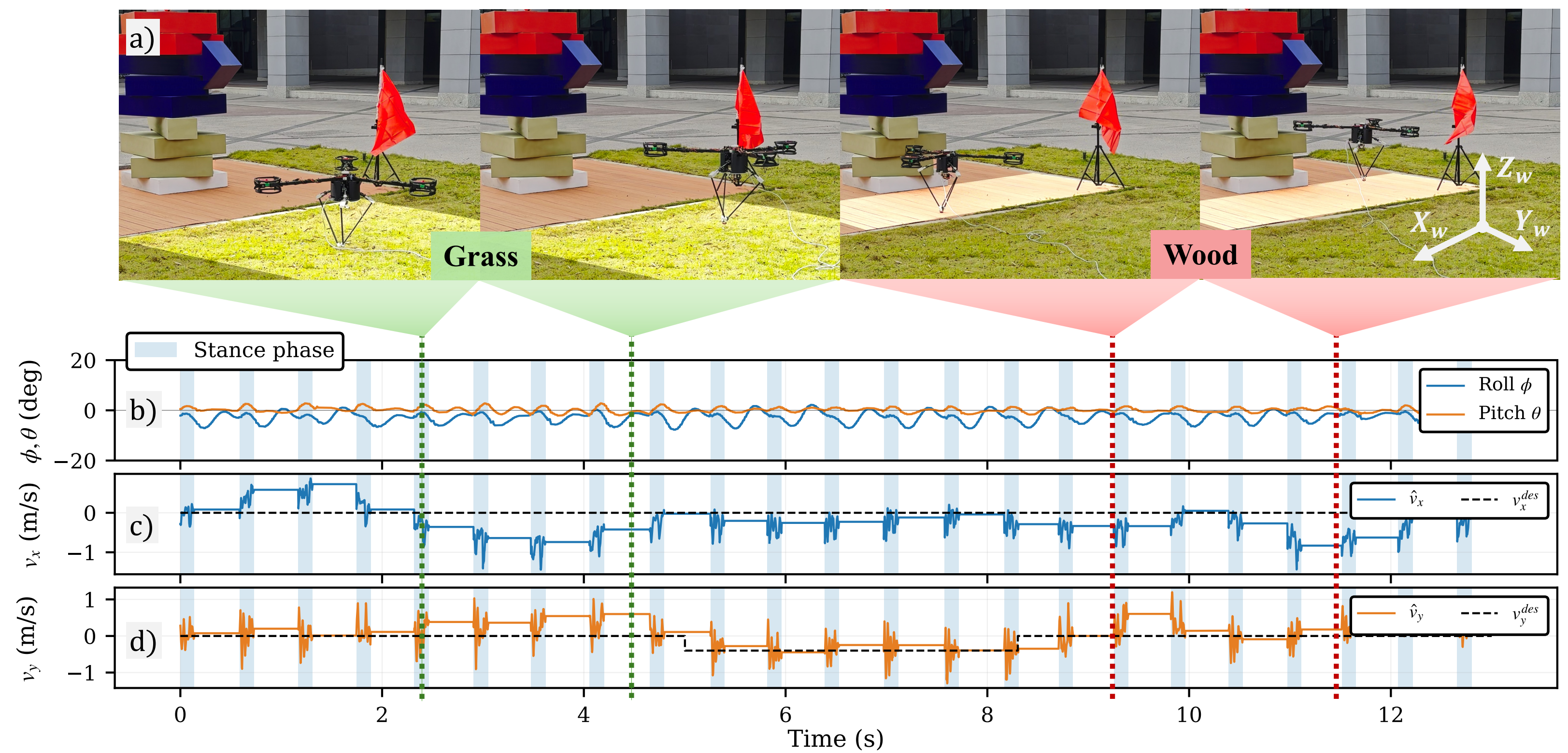
Handoff
Events switch authority; no hardware or gain change.

7. Validation - indoor tracking



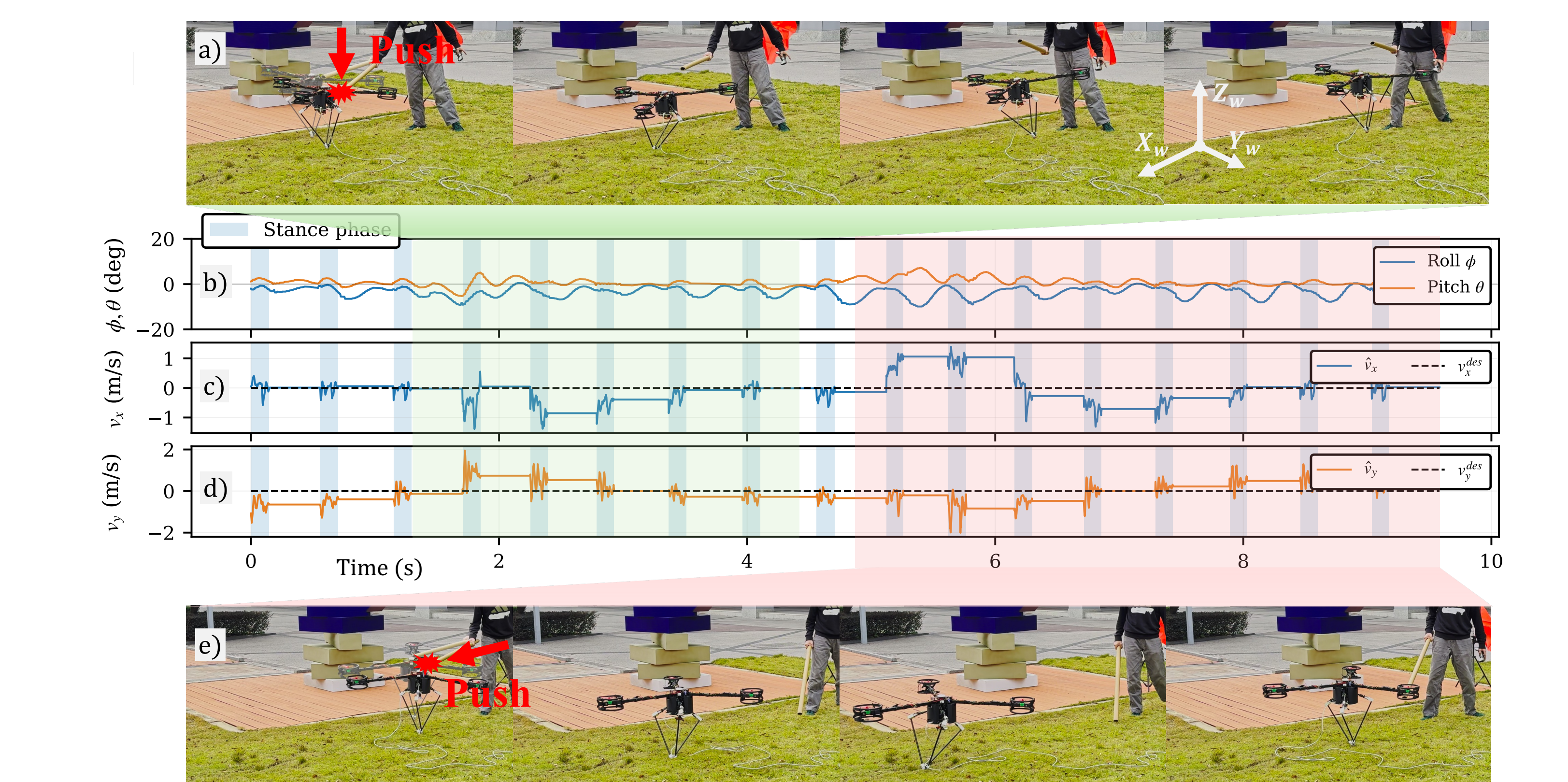
17 hops - 1.6 and 0.94 m/s commands - bounded attitude error through every impact.

8. Validation - grass to wood



23 hops - same controller - no terrain model - no gain retuning.

9. Validation - push recovery



Rotor-arm hit: attitude recovers in 1-2 hops. Main-body hit: lateral velocity recovers in three hops.